S04. Finite-volume certification of a disordered topological index

Pure Thought research portfolio

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Source: synthesis challenge 15; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

A perfectly repeating lattice is convenient for calculating topological properties, but disorder breaks that repetition. This problem asks when a calculation on a finite, disordered patch can prove a topological property of the corresponding infinite system. The key issue is whether the patch is large enough and the model controlled enough for its answer to be trustworthy.

Key terms

- **Hamiltonian** — The operator describing the lattice system's energies and couplings.
- **Disorder** — Spatial variation in parameters that need not repeat periodically.
- **Locality** — A restriction on how far or how strongly sites can interact.
- **Fermi energy** — The reference energy separating occupied and unoccupied states in the specified zero-temperature filling model.
- **Spectral gap** — An open energy interval with no spectrum of the bulk Hamiltonian.
- **Topological index** — A discrete quantity that distinguishes phases and stays fixed under allowed gap-preserving changes.
- **Finite volume** — A bounded patch of a larger or infinite lattice.
- **Spectral localizer** — A matrix combining Hamiltonian and position information to compute an index under stated conditions.
- **Mobility gap** — A regime characterized by localization rather than necessarily an empty spectral interval; it requires different hypotheses.

Why this matters

Topological behavior is interesting partly because it can survive certain imperfections. Establishing that robustness requires methods that work without a perfect crystal and distinguish a finite-sample observation from a bulk conclusion. A rigorous bridge between the two makes numerical evidence more informative.

What progress would mean

Progress would give checkable size, gap and locality conditions under which a finite calculation equals the bulk index. It would not automatically establish robustness to every disorder strength, a mobility-gap theorem, or the transport response of an experimental device.

Revised core question

For a local disordered lattice Hamiltonian with a specified bulk spectral-gap hypothesis, when does a finite-volume index provably equal its infinite-volume counterpart?

Precise scope, target and outputs

Fix a finite-range two-dimensional tight-binding Hamiltonian on Z^2, bounded disorder, Fermi energy E_F and a bulk spectral gap. Initially use bounded perturbations small enough to preserve a known clean gap. Choose one index construction, such as a spectral localizer, and one theorem relating finite and infinite volume. Define position operators, truncation radius and boundary convention; compute the finite Hermitian signature or index with interval spectral bounds. Verify every locality, commutator, radius and scale inequality in the selected theorem. Mobility-gap cases require a different theorem and are an extension.

Detailed critique of the synthesis framing

The synthesis lists Bott indices, noncommutative Chern numbers and localizers as if they shared one universal finite-size error bound. They have different hypotheses and finite-volume behavior. An integer from a finite matrix is not automatically the bulk invariant, and one disorder realization does not establish robustness for all disorder. The revised task certifies the finite-to-infinite transfer under a stated gap/locality theorem; it is distinct from the periodic model classification in PRD 09.

Validation and milestones

Baseline: recover a clean model's known index with a validated finite-to-bulk theorem. Strong: certify a full bounded disorder class using gap preservation. Include trivial controls and a gap-closing case that the checker declines to certify.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$\operatorname{dist}(E_F, \operatorname{spec} H_0) \geq g > 0, \qquad \|V\| < g \quad \Longrightarrow \quad \operatorname{dist}(E_F, \operatorname{spec}(H_0 + V)) \geq g - \|V\|.$$

This bounded self-adjoint perturbation estimate supplies a spectral-gap hypothesis. Equality of a chosen finite-volume index with the bulk index still requires a specific localizer or other index theorem with all radius, locality, commutator and scale conditions checked. Mobility-gap claims need a different theorem.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.